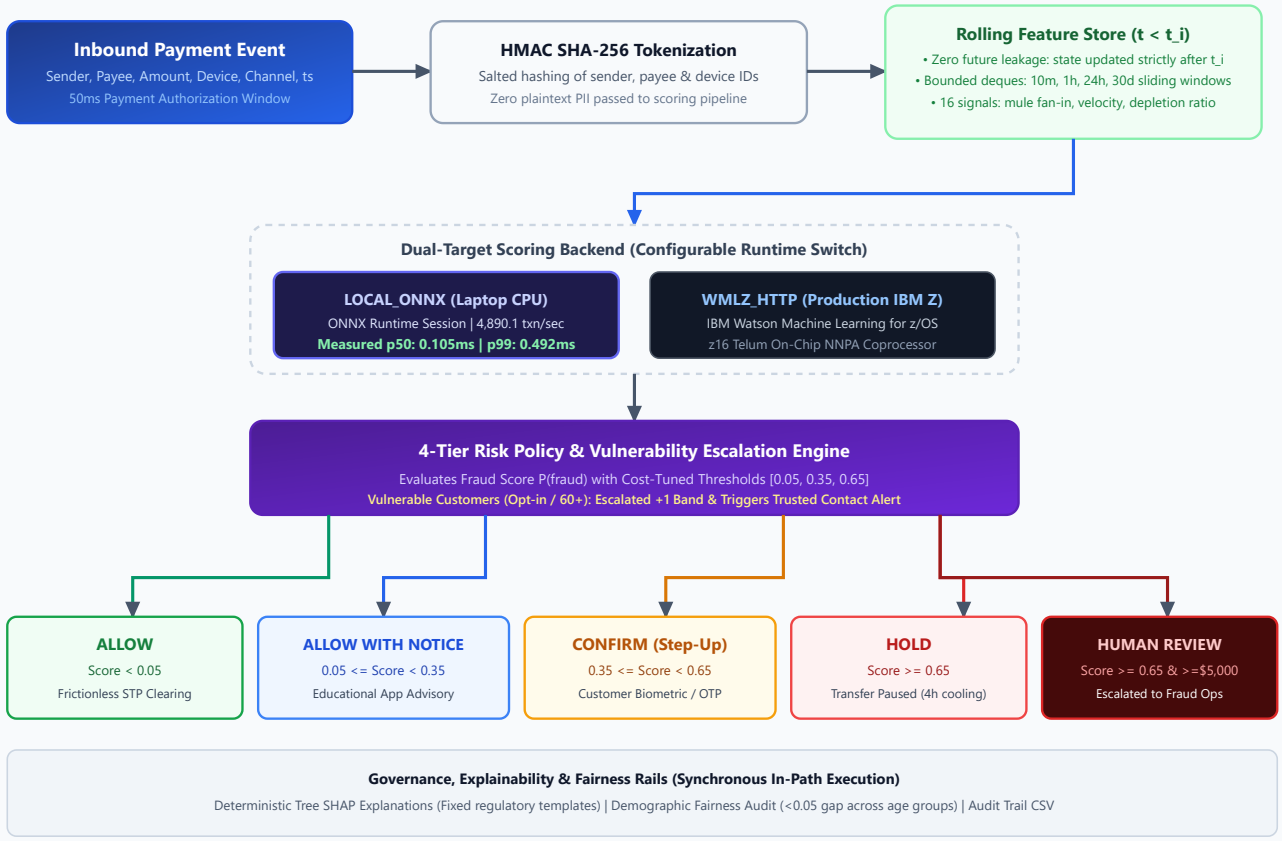
ScamShield Z — Solution Flowchart

Real-Time AI Scam & Authorized Push Payment Fraud Detection Architecture

IBM Z Datathon 2026
Theme: Real-Time AI for Critical Decisions

VERIFIED P99 = 0.492 MS

End-to-End Transaction Decisioning Flowchart



Engine & Performance Specifications

Scoring Engine:	`ONNX Runtime (CPU) & WMLz HTTP`
Verified p50 Latency:	`0.105 ms (104.6 µs)`
Verified p95 Latency:	`0.320 ms (319.5 µs)`
Verified p99 Latency:	`0.492 ms (491.7 µs)`
Sustained Throughput:	`4,890.1 transactions / sec`
ONNX Max Parity Diff:	`5.93e-08 (< 1e-05 pass)`

Risk Policy & Cost Matrix Optimization

Low Risk (STP Allow):	`P(fraud) < 0.05`
Medium Risk (Notice / Confirm):	`0.05 <= P < 0.35`
High Risk (Confirm / Hold):	`0.35 <= P < 0.65`
Very High Risk (Hold / Review):	`P >= 0.65`
Human Review Threshold:	`Amount >= $5,000.00`
Cost Optimization Matrix:	`Miss: 50 | Hold: 5 | Prompt: 1`

Model Evaluation & Data Provenance Summary

Model / Dataset	Data Type	PR-AUC	ROC-AUC	Recall @ 1% FPR	Role in Verification
LightGBM (ONNX Runtime)	Synthetic Stream (200k)	0.9994	1.0000	1.0000	Pipeline integrity & zero future-leakage verification
Baseline Logistic Regression	Synthetic Stream (200k)	0.8636	0.9992	0.9833	Linear classification baseline
ULB Credit Card Fraud (Real Data)	Authentic European Cards (100k)	**0.2803**	**0.8857**	**0.7708**	**Real-world baseline with authentic noisy fraud**

Code Repository: github.com/AVJaswanth/ScamShield-Z
License: Apache 2.0 / MIT